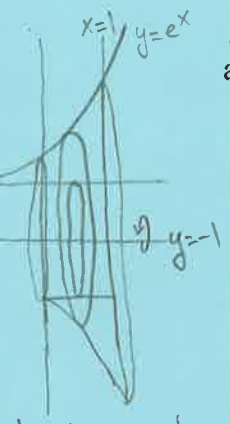
* Note: As Janita made this answer key, the problems are worked out as she thinks about them. You may think about it differently, but your integral and answer will be the same.

MATH 152 **QUIZ 2A** Name Answer Key

Mrs. Bonny Tighe 6.2-6.3

25 points TA_____ 9/13/17

1. Find the volume of the solid formed by rotating the region bounded by $y = e^x$, $x = 0$, $y = 0$ *and* $x = 1$ about the line y = - 1. Sketch the region and the solid and a representative **disc or washer**.



Limits of integration: I'm given that my region is bounded by x=0 and x=1 so my limits of integration are x=0,1

As the washers are stacked next to each other, this is a dx problem

$R_{out} = e^x - (-1) = e^x + 1$

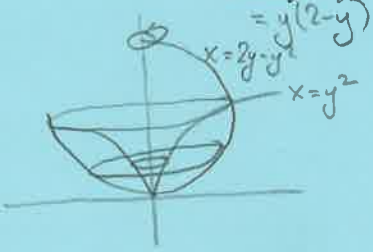
$R_{in} = 0 - (-1) = 1$

$$V = \pi\int_0^1 \left[R_{out}^2 - R_{in}^2\right]dx$$
$$= \pi\int_0^1 \left[(e^x+1)^2 - 1^2\right]dx$$
$$= \pi\int_0^1 \left[e^{2x} + 2e^x + 1 - 1\right]dx$$
$$= \pi\int_0^1 \left[e^{2x} + 2e^x\right]dx$$
$$= \pi\left(\tfrac{1}{2}e^{2x} + 2e^x\right|_0^1$$
$$= \pi\left(\tfrac{1}{2}e^2 + 2e - \tfrac{1}{2}e^0 - 2e^0\right)$$
$$= \boxed{\pi\left(\tfrac{1}{2}e^2 + 2e - \tfrac{5}{2}\right)}$$

3. Find the volume of the solid formed by rotating the given region about the specified line. Sketch the region and the solid and a representative **disc or washer**.

$x = 2y - y^2$ *and* $y^2 = x$ about the ~~x~~ y-axis.

$= y(2-y)$



As the washers are stacked on top of each other, this is a dy problem.

Points of intersection:

$y^2 = 2y - y^2$

$2y^2 - 2y = 0$

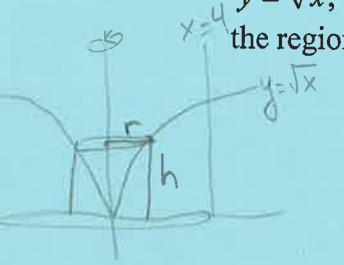
$2y(y-1) = 0$

$y = 0, 1$

$R_{out} = 2y - y^2$, $R_{in} = y^2$

$$V = \pi\int_0^1 \left[R_{out}^2 - R_{in}^2\right]dy$$
$$= \pi\int_0^1 \left[(2y-y^2)^2 - (y^2)^2\right]dy$$
$$= \pi\int_0^1 \left[4y^2 - 4y^3 + y^4 - y^4\right]dy$$
$$= \pi\int_0^1 \left[4y^2 - 4y^3\right]dy$$
$$= \pi\left(\tfrac{4}{3}y^3 - y^4\right|_0^1$$
$$= \pi\left[\tfrac{4}{3}(1)^3 - 1^4 - 0\right]$$
$$= \pi\left[\tfrac{4}{3} - 1\right]$$
$$= \boxed{\frac{\pi}{3}}$$

3. Find the volume of the solid obtained by rotating the region bounded by $y=\sqrt{x}$, $x=4$ *and* $y=0$, about the y-axis using the **cylindrical shells** method. Sketch the region and a representative cylinder. .



$r = x, \quad h = \sqrt{x}$

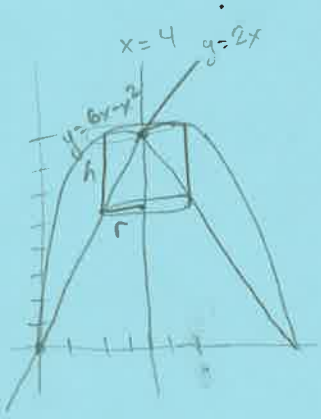
$$V = 2\pi \int_0^4 x\sqrt{x}\,dx$$
$$= 2\pi \int_0^4 x^{3/2}\,dx$$
$$= 2\pi \left(\tfrac{2}{5} x^{5/2}\right|_0^4$$
$$= 2\pi\left(\tfrac{2}{5}\right)\left[(4)^{5/2} - 0\right]$$
$$= \frac{4\pi}{5}\left(2^5\right)$$
$$= \boxed{\frac{128\pi}{5}}$$

My shells are building outwards (in the x direction), so this is a dx problem.

Limits of integration: Note we start at $x=0$ and are bounded by $x=4$, so our limits are $x = 0, 4$

4. Use the cylindrical shells method to find the volume of the solid of revolution obtained by rotating the region bounded by $y = 6x - x^2$ *and* $y = 2x$ about the line x = 4. Sketch the graph

$= x(6-x)$



$r = 4 - x, \quad h = 6x - x^2 - 2x = 4x - x^2$

$$V = 2\pi \int_0^4 r h\,dx$$
$$= 2\pi \int_0^4 (4-x)(4x-x^2)\,dx$$
$$= 2\pi \int_0^4 \left[16x - 4x^2 - 4x^2 + x^3\right] dx$$
$$= 2\pi \int_0^4 \left[x^3 - 8x^2 + 16x\right] dx$$
$$= 2\pi \left(\tfrac{1}{4}x^4 - \tfrac{8}{3}x^3 + 8x^2\right|_0^4$$
$$= 2\pi\left[\tfrac{1}{4}(4)^4 - \tfrac{8}{3}(4)^3 + 8(4)^2 - 0\right]$$
$$= 2(4^3)\pi\left[1 - \tfrac{8}{3} + 2\right]$$
$$= 128\pi\left(3 - \tfrac{8}{3}\right)$$
$$= \boxed{\frac{128\pi}{3}}$$

Again my shells are building in the x direction, so this problem is a dx problem.

Points of intersection:

$6x - x^2 = 2x$

$0 = x^2 - 4x$

$0 = x(x-4)$

$x = 0, 4$

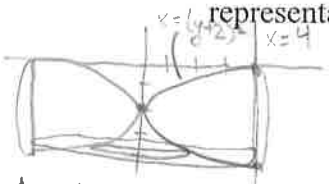
* Note: As Janita made this answer key, the problems are worked out as she thinks about them. You may think about it differently, but your integral and answer will be the same.

MATH 152 **QUIZ 2** Name Answer Key

Mrs. Bonny Tighe 6.2-6.3

25 points TA_____ 9/13/17

1. Find the volume of the solid formed by rotating the region bounded by $x = (y+2)^2$ *and* $x = 4$ about the y-axis. Sketch the region and the solid and a representative **disc or washer**.



As the washers are stacked on top of each other, this is a dy problem.

Points of intersection:

$(y+2)^2 = 4$

$y+2 = \pm 2$

$y+2 = 2$ or $y+2 = -2$

$y = 0$ $\quad$ $y = -4$

$R_{out} = 4$, $R_{in} = (y+2)^2$

$$V = \pi \int_{-4}^{0} \left[R_{out}^2 - R_{in}^2\right] dy$$

$$= \pi \int_{-4}^{0} \left[4^2 - ((y+2)^2)^2\right] dy$$

$$= \pi \int_{-4}^{0} \left[16 - (y+2)^4\right] dy$$

$$= \pi \left[\int_{-4}^{0} 16\, dy - \int_{-4}^{0} (y+2)^4 dy\right] \quad u = y+2,\ du = dy$$

$$= \pi \left[\left(16y\right|_{-4}^{0} - \int u^4 du\right]$$

$$= \pi \left[0 - (16)(-4) - \left(\tfrac{1}{5}u^5\right|\right]$$

$$= \pi \left[64 - \left(\tfrac{1}{5}\right)(y+2)^5\Big|_{-4}^{0}\right]$$

$$= \pi \left[64 - \tfrac{1}{5}(2)^5 + \tfrac{1}{5}(-2)^5\right]$$

$$= \pi \left[64 - \tfrac{32}{5} - \tfrac{32}{5}\right]$$

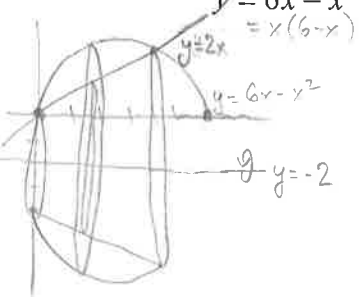
$$= \pi \left[64 - \tfrac{64}{5}\right]$$

$$= 64\pi \left[1 - \tfrac{1}{5}\right]$$

$$= 64\pi \left(\tfrac{4}{5}\right) = \boxed{\frac{256\pi}{5}}$$

3. Find the volume of the solid formed by rotating the given region about the specified line. Sketch the region and the solid and a representative **disc or washer**.

$y = 6x - x^2$ *and* $y = 2x$ about the line $y = -2$



As the washers are stacked next to each other, this is a dx problem.

Points of intersection:

$6x - x^2 = 2x$

$0 = x^2 - 4x$

$0 = x(x-4)$

$x = 0, 4$

$R_{out} = 6x - x^2 + 2$, $R_{in} = 2x + 2$

$$V = \pi \int_{0}^{4} \left[R_{out}^2 - R_{in}^2\right] dx$$

$$= \pi \int_{0}^{4} \left[(6x - x^2 + 2)^2 - (2x+2)^2\right] dx$$

$$= \pi \int_{0}^{4} \left[36x^2 + x^4 + 4 - 12x^3 - 4x^2 + 24x - 4x^2 - 8x - 4\right] dx$$

$$= \pi \int_{0}^{4} \left[x^4 - 12x^3 + 28x^2 + 16x\right] dx$$

$$= \pi \left(\tfrac{1}{5}x^5 - 3x^4 + \tfrac{28}{3}x^3 + 8x^2\Big|_{0}^{4}\right.$$

$$= \pi \left[\tfrac{1}{5}(4)^5 - 3(4)^4 + \tfrac{28}{3}(4)^3 + 8(4)^2 - 0\right]$$

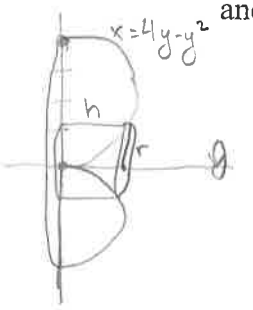
$$= 4^3\pi \left[\tfrac{16}{5} - 12 + \tfrac{28}{3} + 2\right]$$

$$= 64\pi \left[\frac{48 + 140}{15} - 10\right]$$

$$= 64\pi \left[\frac{188 - 150}{15}\right]$$

$$= \boxed{\frac{2432\pi}{15}}$$

3. Find the volume of the solid obtained by rotating the region bounded by $x = 4y - y^2 = y(4-y)$ and the y-axis, about the x-axis using the **cylindrical shells** method. Sketch the region and a representative cylinder. .



$r = y, \; h = 4y - y^2$

$$V = 2\pi \int_0^4 r h \, dy$$
$$= 2\pi \int_0^4 y(4y - y^2)\,dy$$
$$= 2\pi \int_0^4 (4y^2 - y^3)\,dy$$
$$= 2\pi \left(\tfrac{4}{3}y^3 - \tfrac{1}{4}y^4\right|_0^4$$
$$= 2\pi \left[\tfrac{4}{3}(4)^3 - \tfrac{1}{\cancel{4}}(4)^{\cancel{4}\,3} - 0\right]$$
$$= 2(4^3)\pi\left[\tfrac{4}{3} - 1\right]$$
$$= 128\pi\left(\tfrac{1}{3}\right)$$
$$= \boxed{\frac{128\pi}{3}}$$

My shells are being built in the y direction, so this is a dy problem.

Points of intersection

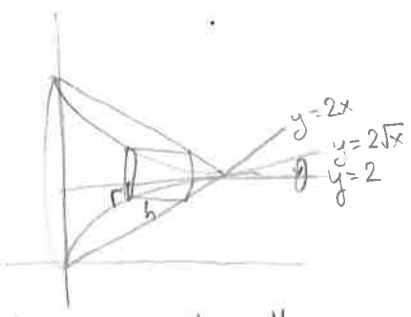
$0 = 4y - y^2$

$0 = y(4-y)$

$y = 0, 4$

4. Use the cylindrical shells method to find the volume of the solid of revolution obtained by rotating the region bounded by $y = 2\sqrt{x}$ *and* $y = 2x$ about the line y = 2. Sketch the graph

$x = \frac{y^2}{4}$ $\quad$ $x = \frac{1}{2}y$



$r = 2 - y, \; h = \frac{1}{2}y - \frac{1}{4}y^2$

$$V = 2\pi \int_0^2 r h \, dy$$
$$= 2\pi \int_0^2 (2-y)\left(\tfrac{1}{2}y - \tfrac{1}{4}y^2\right)dy$$
$$= 2\pi \int_0^2 \left[y - \tfrac{1}{2}y^2 - \tfrac{1}{2}y^2 + \tfrac{1}{4}y^3\right]dy$$
$$= 2\pi \int_0^2 \left[\tfrac{1}{4}y^3 - y^2 + y\right]dy$$
$$= 2\pi \left(\tfrac{1}{16}y^4 - \tfrac{1}{3}y^3 + \tfrac{1}{2}y^2\right|_0^2$$
$$= 2\pi \left[\tfrac{1}{\cancel{16}}(\cancel{2})^{\cancel{4}} - \tfrac{1}{3}(2)^3 + \tfrac{1}{\cancel{2}}(2)^{\cancel{2}} - 0\right]$$
$$= 2\pi\left[1 - \tfrac{8}{3} + 2\right]$$
$$= 2\pi\left[3 - \tfrac{8}{3}\right]$$
$$= \boxed{\frac{2\pi}{3}}$$

Again, my shells will be built in the y direction, so this is a dy problem.

Points of intersection

$2\sqrt{x} = 2x$

$\sqrt{x} = x$

$x = x^2$

$0 = x^2 - x$

$0 = x(x-1)$

$x = 0, 1$

$\Rightarrow y = 0, 2$